Event Driven State Machine Lab

BME554L - Fall 2026 - Palmeri

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State Specifications

The final project for this class will be a heart rate and temperature monitor that will send metrics to a mobile phone using Bluetooth Low Energy (BLE). The device states are described in ecg-temp-ble-lab.qmd.

This assigment will be the starting point to get feedback on creating a complete and descriptive state diagram for the device.

Generate the State Diagram

> **Tip**
>
> - For each state in your diagram, consider if a state would benefit from having entry/exit substates.
> - Err on the side of more detail than not at this stage.

1. Generate a hand-drawn (electronic or scan of paper) state diagram for the device. Include all states, transitions, and events that trigger transitions. Include any entry/exit actions for each state.

2. Electronically render your state diagram using one of the tools outlined on the State Diagram Tools page.

> **Tip**
>
> For UML digrams, if the diagram's layout is difficult to follow using the automated layout, consider manually specifying relative positions and length of arrows using the following annotation guidelines: https://mermaid.ai/open-source/syntax/stateDiagram.html.

Submission

Upload both renderings of your state diagram to the associated Canvas assignment.

Expectations

A "perfect" state diagram should not be the expectation for this assignment; constructive feedback on a thoughtful first-pass at a state diagram is the goal. The state diagram will be used to guide the implementation of the device in future labs.